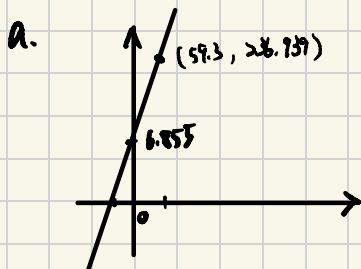


- 3.18 A life insurance company examines the relationship between the amount of life insurance held by a household and household income. Let $INCOME$ be household income (thousands of dollars) and $INSURANCE$ the amount of life insurance held (thousands of dollars). Using a random sample of $N = 20$ households, the least squares estimated relationship is

$$\widehat{INSURANCE} = 6.855 + 3.880INCOME$$

(se) (7.383) (0.112)

- Draw a sketch of the fitted relationship identifying the estimated slope and intercept. The sample mean of $INCOME = 59.3$. What is the sample mean of the amount of insurance held? Locate the point of the means in your sketch.
- How much do we estimate that the average amount of insurance held changes with each additional \$1000 of household income? Provide both a point estimate and a 95% interval estimate. Explain the interval estimate to a group of stockholders in the insurance company.



$$\begin{aligned}\widehat{insurance} &= 6.855 + 3.88 \cdot 59.3 \\ &= 28.939\end{aligned}$$

- b.
- ① When household income increases \$1000, the average amount of insurance increases 3.88 units.

$$\textcircled{2} \quad df = 20 - 2 = 18$$

$$b_2 \pm t_{0.025}(18) \text{ se}(b_2)$$

$$= 3.88 \pm 2.101 \cdot 0.112$$

$$= 3.645 - 4.115$$

$$\Rightarrow 95\% \text{ C.I. } [3.645, 4.115]$$

We are 95% confident that with each additional \$1000 of income, the insurance held by household will be within $[3.645, 4.115]$

- c. Construct a 99% interval estimate of the expected amount of insurance held by a household with \$100,000 income. The estimated covariance between the intercept and slope coefficient is -0.746 .

(99)

$$\begin{aligned}\text{Var}(\widehat{\text{insurance}}) &= \text{Var}(b_1) + \text{INCOME}^2 \text{Var}(b_2) + 2 \cdot \text{INCOME} \text{Cov}(b_1, b_2) \\ &= 7.383^2 + 100^2 \cdot 0.112^2 + 200 \cdot (-0.746) \\ &= 20.749\end{aligned}$$

$$\text{SE}(\widehat{\text{insurance}}) = \sqrt{20.749} = 5.544$$

$$\widehat{\text{insurance}} = 6.855 + 3.88 \cdot 100 = 394.855$$

99% C.I.:

$$\begin{aligned}& [\widehat{\text{insurance}} \pm t_{0.005}(18) \text{SE}(\widehat{\text{insurance}})] \\ &= [378.896, 410.814]\end{aligned}$$

- d. One member of the management board claims that for every \$1000 increase in income the average amount of life insurance held will increase by \$5000. Let the algebraic model be $\text{INSURANCE} = \beta_1 + \beta_2 \text{INCOME} + e$. Test the hypothesis that the statement is true against the alternative that it is not true. State the conjecture in terms of a null and alternative hypothesis about the model parameters. Use the 5% level of significance. Do the data support the claim or not? Clearly, indicate the test statistic used and the rejection region.

$$\begin{aligned}\textcircled{1} H_0: b_2 &= 5 \\ H_a: b_2 &\neq 5\end{aligned}$$

$$\textcircled{2} \alpha = 0.05$$

$$\textcircled{3} T = \frac{b_2 - 5}{\text{SE}(b_2)} \sim_{H_0} t(20.2)$$

$$\textcircled{4} \text{RR} = \{ |T| \geq t_{0.025}(18) = 2.101 \}$$

$$\textcircled{5} T_0 = \frac{3.88 - 5}{0.12} = -10$$

$$\textcircled{6} \because T_0 \in \text{RR}$$

$\Rightarrow$ reject H_0 , the

data don't support the claim.

- e. Test the hypothesis that as income increases the amount of life insurance held increases by the same amount. That is, test the null hypothesis that the slope is one. Use as the alternative that the slope is larger than one. State the null and alternative hypotheses in terms of the model parameters. Carry out the test at the 1% level of significance. Clearly indicate the test statistic used, and the rejection region. What is your conclusion?

$$\textcircled{1} H_0 = b_2 = 1$$

$$H_a: b_2 \neq 1$$

$$\textcircled{2} \alpha = 0.01$$

$$\textcircled{3} T = \frac{b_2 - 1}{SE(b_2)} \xrightarrow{H_0} t_{(20-2)}$$

$$\textcircled{4} RR = \{ |T| \geq t_{0.005}(18) = 2.878 \}$$

$\textcircled{5}$

$$T_0 = \frac{3.88 - 1}{0.112} = 25.714$$

$\textcircled{6}$

$\therefore T_0 \in RR$, reject H_0

$\Rightarrow$ the slope is significantly not equal to 1

33.

a.

=====

值

t 統計量 -26.7287516645277

臨界值 1.64791913885501

p 值 1

結論 無法拒絕 H0

There is no statistical evidence to suggest that the marginal effect of increasing house size by 100 square feet (for a 2000 sqft house) is greater than \$13,000.

b.

=====

值

t 統計量 4.18946671475947

臨界值 1.64791913885501

p 值 1.65394980079715e-05

結論 拒絕 H0

There is strong statistical evidence to conclude that the marginal effect on the expected house price of increasing the size of a 2000 square foot house by 100 square feet is greater than \$13,000.

c.

=====

值

點估計值 167.373

標準誤差 4.746

t 臨界值 1.965

95% 下界 158.048

95% 上界 176.699

The point estimate for the expected price of a house with 2000 square feet is

167.373. Based on the model, the 95% confidence interval for the mean expected price is [158.048, 176.699]. This implies that we are 95% confident that the true population average price for houses of this specific size falls within this range.

d.

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值

mean price 163.333

The sample average of 163.333 is compatible with the result in part (c).

31.

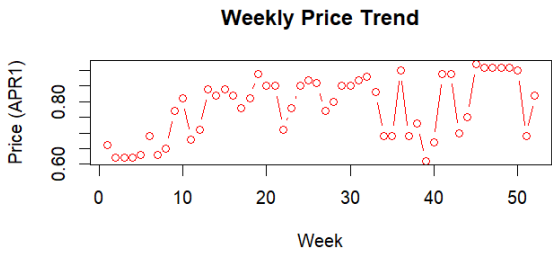
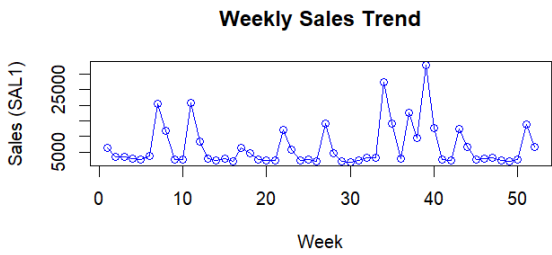
a.

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Statistic	Mean	Min	Max	St. Dev.
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sal1	6,718.712	1,762	32,820	6,915.083
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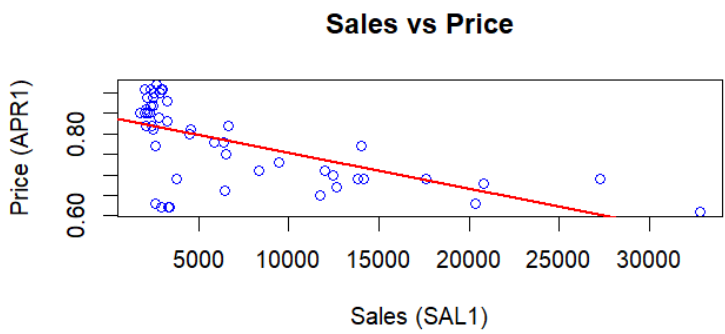
apr1	0.782	0.610	0.920	0.098
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The visual analysis reveals significant week-to-week variation in both variables. SAL1 exhibits highly volatile behavior characterized by dramatic spikes. Most importantly,

there is an inverse relationship between the two: every sharp increase in sales perfectly coincides with a significant price drop.

b.



The chart confirms that sales tend to drop off when prices go up. This is in line with our expectations and reflects the basic law of demand.

c.

=====	
Dependent variable:	

	sal1

price	-434.447*** (78.585)
Constant	40,714.220*** (6,196.421)

Observations	52
R2	0.379
Adjusted R2	0.367
Residual Std. Error	5,501.925 (df = 50)
F Statistic	30.563*** (df = 1; 50)
=====	
Note:	*p<0.1; **p<0.05; ***p<0.01
=====	

值

Lower bound -592.290
upper bound -276.605

For every one-cent increase in unit price, the weekly sales are expected to contract by 434.447 units.

d.

=====

值

Predicted Sales 10,302.900
90% Lower 8,624.934
90% Upper 11,980.870

With an absolute elasticity exceeding 1, the demand is highly elastic. This high sensitivity is expected in the competitive canned tuna market, where low switching costs and a high density of rival brands allow consumers to react strongly to even minor price changes.

e.

=====

值

es -5.060
Lower -6.898
Upper -3.222

With an absolute elasticity exceeding 1, the demand is highly elastic. This high sensitivity is expected in the competitive canned tuna market, where low switching costs and a high density of rival brands allow consumers to react strongly to even minor price changes.

f.

=====

值

t-stat -2.25057192191693

tc 1.6759050251631

p-value 0.0288420363284799

conclusion 拒絕 H0

Since the t-statistic exceeds the critical value and the p-value is less than 0.10, we reject the null hypothesis. Consequently, there is sufficient evidence to conclude that the price elasticity of sales for Brand No. 1 canned tuna is significantly different from -3 at the 10% significance level.